

Week # 01Lecture # 01MultivariableCalculus

- Calculus means change.

Book Recommendation :

- Functions

Thomas Calculus

- For Single Variable

- For Multivariable

$$f(x) = 2x \rightarrow \text{Dependent}$$

$\rightarrow$ Independent

$$f(x, y) = 2x + 7y$$

 $\Rightarrow$ Limits and Continuity

Approaching value used in limits.

 $\Rightarrow$ Single variable limiting Functions :

$$\lim_{x \rightarrow 10} f(x) = 2x$$

 $\Rightarrow$ Multi Variable limiting Functions :

$$\text{i. } \lim_{(x, y) \rightarrow (1, 2)} f(x, y) = \lim_{(x, y) \rightarrow (1, 2)} (2x + 3y)$$

$$\text{ii. } \lim_{(x, y, z) \rightarrow (1, 2, 3)} f(x, y, z) = \lim_{(x, y, z) \rightarrow (1, 2, 3)} (2x + 3y + 4z)$$

Questions

1. $\lim_{(x,y) \rightarrow (0,1)} \frac{x - xy + 3}{x^2y + 5xy - y^3}$

Sol. $= \frac{0 - (0)(1) + 3}{(0^2)(1) + 5(0)(1) - (1)^3} = \frac{0 - 0 + 3}{0 + 0 - 1} = -\frac{3}{1} = -3 \text{ Ans.}$

2. $\lim_{(x,y) \rightarrow (3,-4)} \sqrt{x^2 + y^2}$

Sol. $= \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5 \text{ Ans.}$

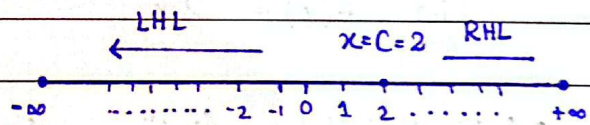
Lecture # 02

Continuity of Functions

$f(x)$

$f_2(x)$ $x = c = 2$ $f_3(x)$

$f_1(x)$



$f(x) = \begin{cases} f_1(x), & x \neq 2 \\ f_2(x), & x < 2 \\ f_3(x), & x > 2 \end{cases}$

Piecewise
Function

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M T W T F S S

Properties :-

1. $f(c)$ defined (Functional value defined).

2. $\lim_{x \rightarrow c} f(x)$ Exists.

$\Rightarrow$ Left hand limit
 $\lim_{x \rightarrow c^-} f(x) \neq \text{LHL}$

$\Rightarrow$ Right hand limit
 $\lim_{x \rightarrow c^+} f(x)$

$\Rightarrow$ If $\cdot$ LHL = RHL then limit exist.

3. $f(c) = \lim_{x \rightarrow c} f(x)$

For Single Variable

Questions

$$1. \quad f(x) = \begin{cases} 3x & , \quad x = 2 \\ 3x+4 & , \quad x < 2 \\ 4x+4 & , \quad x > 2 \end{cases} \quad \text{at } x = c = 2$$

1 - $f(2) = 3x = 3(2) = 6$

2. LHL

$$= \lim_{x \rightarrow 2^-} 3x+4$$

$$= 3(2)+4 = 10$$

RHL

$$= \lim_{x \rightarrow 2^+} 4x+4$$

$$= 4(2)+4 = 12$$

Result: $LHL \neq RHL$, So function is discontinuous at $x = c$

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2. 1) Discuss the continuity at point $x=c=2$.

$$f(x) = \begin{cases} 2x+5 & \text{if } x \leq 2 \\ 4x+1 & \text{if } x > 2 \end{cases}$$

1. $f(2) = 2x+5 = 2(2)+5$
 $= 4+5 \therefore = 9$

2.

LHL

$$\lim_{x \rightarrow 2^-} 2x+5$$

$$= 2(2)+5$$

$$= 9$$

RHL

$$\lim_{x \rightarrow 2^+} 4x+1$$

$$= 4(2)+1$$

$$= 9$$

3. $f(2) = \lim_{x \rightarrow 2} f(x)$

Result :

• Function $F(x)$ is Continuous at $x=c$.

For Multi-Variable

Q#03:

$$f(x,y) = \begin{cases} \frac{x^2 y}{x^2 + y^2} & , (x,y) \neq (0,0) \\ 0 & , (x,y) = (0,0) \end{cases}$$

$$y = x$$

$$f(x, x) = \frac{x^3}{x^2 + x^2} = \frac{x^3}{2x^2}$$

$$f(x, x) = \frac{x}{2} \text{ when } y$$

along 'x'

ii. LHL

RHL

$$\lim_{(x,x) \rightarrow (0,0)} f(0,0) = \frac{0}{2} = 0$$

$$\lim_{(x,y) \rightarrow (0,0)^+} f(x,y) = \frac{0}{2} = 0$$

$$\underline{LHL = RHL}$$

iii. $f(0,0) = \lim_{(x,x) \rightarrow (0,0)} \frac{x}{2}$

$$0 = 0$$

Function is continuous at $y = x$

Derivative :-

⇒ Rate of Change

change $\leftarrow \frac{df(x)}{dx} \rightarrow$ Dependent
Independent

⇒ Derivative is a instantaneous rate of change.